

- 5 Use of data: Learners are expected to have explored different data sets, using appropriate technology, during the course. No particular data set is expected to be studied, and there will not be any pre-release data.

Hypothesis Tests

Hypotheses should be stated in terms of parameter values (where relevant) and the meanings of symbols should be stated. For example,

" $H_0: p = 0.7$, $H_1: p < 0.7$, where p is the population proportion in favour of the resolution".

Conclusions should be stated in such a way as to reflect the fact that they are not certain. For example, "There is evidence at the 5% level to reject H_0 . It is likely that the mean mass is less than 500 g." "There is no evidence at the 2% level to reject H_0 . There is no reason to suppose that the mean journey time has changed."

Some examples of incorrect conclusion are as follows:

" H_0 is rejected. Waiting times have increased."

"Accept H_0 . Plants in this area have the same height as plants in other areas."

Content of Statistics (Optional paper Y542)

When this course is being co-taught with AS Level Further Mathematics A (H235) the 'Stage 1' column indicates the common content between the two specifications and the 'Stage 2' column indicates content which is particular to this specification. In each section the OCR reference lists 'Stage 1' statements before 'Stage 2' statements.

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
5.01 Probability			
5.01a	Probability	a) Be able to evaluate probabilities by calculation using permutations and combinations. <i>Includes the terms "permutation" and "combination".</i> <i>Includes the notation ${}_nP_r = {}^nP_r$ and ${}_nC_r = {}^nC_r$.</i> <i>For underlying content on probability see H240 section 2.03.</i>	
5.01b		b) Be able to evaluate probabilities by calculation in contexts involving selections and arrangements. <i>Selection problems include, for example, finding the probability that 3 vowels and 2 consonants are chosen when 5 letters are chosen at random from the word 'CALCULATOR'.</i> <i>Arrangement problems only involve arrangement of objects in a line and include:</i> <i>repetition, e.g. the probability that the word 'ARTIST' is formed when the letters of the word 'STRAIT' are chosen at random.</i> <i>restriction, e.g. the probability that two consonants are (or are not) next to each other when the letters of the word 'TRAITS' are placed in a random order.</i>	

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5.02 Discrete Random Variables			
5.02a	Probability distributions for general discrete random variables	a) Understand and be able to use discrete probability distributions. <i>Includes using and constructing probability distribution tables and functions relating to a given situation involving a discrete random variable.</i> <i>Any defined non-standard distribution will be finite.</i>	
5.02b		b) Understand and be able to calculate the expectation and variance of a discrete random variable. <i>Includes knowing and being able to use the formulae</i> $\mu = E(X) = \sum x_i p_i$ $\sigma^2 = \text{Var}(X) = \sum (x_i - \mu)^2 p_i = \sum x_i^2 p_i - \mu^2.$ <i>[Proof of these results is excluded.]</i>	
5.02c		c) Know and be able to use the effects of linear coding on the mean and variance of a random variable.	
5.02d	The binomial distribution	d) Know and be able to use the formulae $\mu = np$ and $\sigma^2 = np(1 - p)$ for a binomial distribution. <i>[Proof of these results is excluded.]</i> <i>For the underlying content on binomial distributions, see H240 sections 2.04b and 2.04c.</i>	
5.02e	The discrete uniform distribution	e) Know and be able to use the conditions under which a random variable will have a discrete uniform distribution, and be able to calculate probabilities and the mean and variance for a given discrete uniform distribution. <i>Includes use of the notation $X \sim U(n)$ for the uniform distribution over the interval $[1, n]$.</i>	

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5.02f	The geometric distribution	f) Know and be able to use the conditions under which a random variable will have a geometric distribution. <i>Includes use of the notation $X \sim \text{Geo}(p)$, where X is the number of trials up to and including the first success.</i>	
5.02g		g) Be able to calculate probabilities using the geometric distribution. <i>Learners may use the formulae $P(X = x) = (1 - p)^{x-1}p$ and $P(X > x) = (1 - p)^x$.</i>	
5.02h		h) Know and be able to use the formulae $\mu = \frac{1}{p}$ and $\sigma^2 = \frac{1-p}{p^2}$ for a geometric distribution. <i>[Proof of these results is excluded.]</i>	

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5.02i	The Poisson distribution	i) Understand informally the relevance of the Poisson distribution to the distribution of random events, and be able to use the Poisson distribution as a model. <i>Includes use of the notation $X \sim \text{Po}(\lambda)$, where X is the number of events in a given interval.</i>	
5.02j		j) Understand and be able to use the formula .	
5.02k		$P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$ k) Be able to calculate probabilities using the Poisson distribution, using appropriate calculator functions. <i>Learners are expected to have a calculator with the ability to access probabilities from the Poisson distribution.</i> <i>[Use of the Poisson distribution to calculate numerical approximations for a binomial distribution is excluded.]</i>	
5.02l		l) Know and be able to use the conditions under which a random variable will have a Poisson distribution. <i>Learners will be expected to identify which of the modelling conditions [assumptions] is/are relevant to a given scenario and to explain them in context.</i>	
5.02m		m) Be able to use the result that if $X \sim \text{Po}(\lambda)$ then the mean and variance of X are each equal to λ.	
5.02n		n) Know and be able to use the result that the sum of independent Poisson variables has a Poisson distribution.	

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5.03 Continuous Random Variables			
5.03a	Continuous random variables		a) Understand and be able to use the concept of continuous random variable, a probability density function (p.d.f.) and a cumulative distribution function (c.d.f). <i>Includes the normal, continuous uniform and exponential distributions.</i> <i>Includes understanding informally the link between the exponential and Poisson distributions.</i> <i>Includes knowing and being able to use the formulae for the mean and variance of the continuous uniform and exponential distributions.</i> <i>For the underlying content on normal distributions, see sections 2.04e, 2.04f and 2.04g.</i>
5.03b	Probability density functions		b) Be able to use a probability density function (where defined piecewise) to solve problems in probabilities. <i>Includes knowing and being able to use $\int_{-\infty}^{\infty} f(x) dx = 1$</i>
5.03c			c) Be able to calculate the mean and/or variance of a continuous random variable using the formulae $\mu = E(X) = \int x f(x) dx$ and $\sigma^2 = \text{Var}(X) = \int (x - \mu)^2 f(x) dx = \int x^2 f(x) dx - \mu^2$
5.03d			d) Be able to use the general result $E(g(X)) = \int g(x) f(x) dx$, where $f(x)$ is the probability density function of the continuous random variable X and $g(x)$ is a function of X.

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5.03e	Cumulative distribution functions		e) Be able to find and use a cumulative dist function (including where defined piecev problems involving probabilities. <i>Includes being able to use $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$</i>
5.03f			f) Know and be able to use the relationship probability density function, $f(x)$, and the distribution function, $F(x)$, and use eithe the median, quartiles and other percenti
5.03g			g) Be able to find and use the cumulative di functions of related variables. <i>e.g. Given the c.d.f. of X, find the c.d.f. of Y and p.d.f. of Y where $Y = X^3$.</i>
5.04 Linear Combinations of Random Variables			
5.04a	Linear combinations of any random variables		a) Be able to use the following results, inclu where $a = b = \pm 1$ and/or $c = 0$: 1. $E(aX + bY + c) = aE(X) + bE(Y) + c$ 2. if X and Y are independent then $Var(aX + bY + c) = a^2 Var(X) + b^2 Var(Y)$
5.04b	Linear combinations of normal random variables		b) Be able to use the following results: 1. if X has a normal distribution then $aX + b$ has a normal distribution, 2. if X and Y have independent norma then $aX + bY$ has a normal distribu

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5.05 Hypothesis Tests and Confidence Intervals			
5.05a	The distribution of $\bar{X}$ and the central limit theorem		a) Know that for any randomly and independently selected sample, X_i, of size n taken from a population with mean μ and variance σ^2, then for the sample mean $\bar{X}$: $E(\bar{X}) = \mu$, $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$ and $\bar{X}$ is approximately normally distributed for large n (approximately $n > 25$). <i>[Proof of these results is excluded.]</i>
5.05b	Unbiased estimates of population mean and variance		b) Know that unbiased estimates of the population mean and variance are given by $\frac{\sum x}{n}$ and $\frac{n}{n-1} \left(\frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2 \right)$ respectively. <i>[Proof of these results is excluded.]</i> <i>Only an informal understanding of "unbiased" is required.</i>
5.05c	Using the normal distribution in hypothesis tests		c) Be able to use a normal distribution to carry out a hypothesis test for a population mean in the following cases: - a sample drawn from a normal population with known, given or assumed variance, - a large sample drawn from any population with known, given or assumed variance, - a large sample, drawn from any population with unknown variance.
5.05d	Confidence intervals		d) Be able to use a normal distribution to find a confidence interval for a population mean in the above cases.

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5.06 Chi-squared Tests			
5.06a	Contingency tables	a) Be able to use a chi-squared (χ^2) test with the appropriate number of degrees of freedom to test for independence in a contingency table and interpret the results of such a test. <i>Rows or columns, as appropriate, should be combined so that each expected frequency is at least 5, and Yates' correction should be used in the special case of a 2×2 table.</i> <i>A table of critical values of the χ^2 distribution will be provided.</i> <i>Includes calculation of expected frequencies and contributions to the test statistic.</i> <i>Questions may require candidates to calculate some expected frequencies and contributions to the test statistic, but will not involve lengthy calculations.</i>	
5.06b 5.06c	Fitting a theoretical distribution	b) Be able to fit a theoretical distribution, as prescribed by a given hypothesis involving a given ratio, proportion or discrete uniform distribution, to given data. <i>Questions may require candidates to calculate some expected frequencies, but will not involve lengthy calculations.</i>	c) Extend their knowledge of fitting distributions to other known or given discrete and continuous distributions. <i>Questions may require candidates to calculate some expected frequencies, but will not involve lengthy calculations.</i>
5.06d	Goodness of fit test	d) Be able to use a χ^2 test with the appropriate number of degrees of freedom to carry out the corresponding goodness of fit test. <i>Where necessary, adjacent classes should be combined so that each expected frequency is at least 5.</i> <i>A table of critical values of the χ^2 distribution will be provided.</i>	

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